

Mobile Wireless Broadband Availability Challenge Process for Governmental Entities and Third Parties

Updated 07/31/2025

Governmental entities and third parties can submit on-the-ground mobile speed test data to help challenge and verify coverage data submitted by mobile wireless broadband service providers.

Submitting Speed Test Data Using Testing Software and Hardware

In addition to using the FCC Speed Test app (or another third-party speed test app approved by OET for use in the challenge process) to collect on-the-ground speed test data, governments and other entities may collect and submit mobile speed test data for use in the challenge process through their own software and hardware. Entities that choose to submit mobile speed test data using their own software and hardware must ensure that (1) the entity submits a complete description of the methodology used to collect the data and substantiate the data and methodology through the certification of a qualified engineer or official, and (2) the testing implements collect a defined set of metrics.

Description of Methodology:

Entities submitting mobile speed test data into the BDC using their own software and hardware must provide the following information on their data collection methodology (at the same time they upload their speed test data):

1. A narrative overview of the overall testing solution, including:
 - a description of the test procedures and whether the testing was app-based, software-based, or hardware-based;
 - the type of software and hardware used for the testing; and
 - a URL link to the software and hardware products used (if available).
2. Whether the testing software affects the natural behavior of a tested device, such as by locking the device to a particular radio access technology or spectrum band.
3. Information about the devices used in the testing, including supported radio access technologies, data service plan, and whether the devices have internal or external antennas (and the height of the external antennas and antenna gain, if applicable).
4. Information about:
 - the duration of speed tests and speed test servers, including server type (e.g., iPerf, Ookla) and main configurations between server and client (e.g., SCTP/UDP/TCP);
 - the geographic location(s) of the server(s);
 - the estimated number of servers; and

- provisioned bandwidth capacity of test servers.

Required Speed Test Metrics:

Generally, governmental entities and other third-party challengers must submit on-the-ground speed test data that meets the same metrics and parameters as that submitted by consumers, which may be found in the [Data Specifications for Mobile Speed Test Data](#) . One notable deviation from the data specification document is that a challenger submitting data using its own software and hardware may submit the International Mobile Equipment Identity (IMEI) of the device used to conduct a speed test instead of the timestamp that test measurement data were transmitted to the app developer's servers, as well as the source IP address and port of the device, as measured by the server.

Governmental and other third-party challengers must use a device that is able to interface with drive test software and/or runs on the Android operating system.

When tests are taken in an in-vehicle mobile environment, governmental entities and other third-party challengers must also indicate whether the test was conducted with the antenna outside of the vehicle.

How a Challenge is Created:

The BDC system will aggregate valid speed tests conducted by consumers with those conducted by governmental and third-party challengers to collectively meet the thresholds (geographic, temporal, and numeric-i.e., testing) specified in 47 CFR § 1.7006(e) of the Commission's rules in order to create a

challenge. Mobile providers are required to respond to challenges from governmental and other third-party challengers in the same way that they are required to respond to consumer challenges.